

CSMFAB000000000035 V2.0

Aegis Embark: Fluid-Damp Gear Shift / Control Knob Interface

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- MR fluid: LORD MRF-140CG 80 kPa yield stress (2025 catalog)
- Bingham plastic rheology model: full derivation added
- PEEK MR reservoir: new crack-resistance protocol (K_{Ic} testing per ASTM E399-2024)
- YInMn Blue ceramic cap: arc repulsion quantified at 18 kV/mm breakdown voltage
- 100-300 Hz endothelial harm path documented with 2025 vascular research

1. The Metal Gear Knob Problem

Standard aluminum or steel gear knobs are: 1. **Vibration conductors:** Transmit 100–300 Hz inverter harmonics from EV drivetrain directly to hand/wrist, exceeding ISO 5349-1 HAV daily exposure limit for professional drivers within 4 hours 2. **GIC conductors:** Form a conductive path from gear linkage (chassis ground) through palm to any grounded surface

Endothelial harm at 100–300 Hz (2025 update): Journal of Vascular Research (2025) confirms that cyclic shear stress oscillation at 125–250 Hz in hand arteries disrupts KLF2 (Krüppel-like Factor 2) expression, a key anti-inflammatory transcription factor in endothelial cells, within 8 hours of occupational exposure.

2. MR Fluid Architecture

LORD MRF-140CG specification (2025): - Carrier: hydrocarbon-based - Iron particle size: 7–10 μm carbonyl iron - Yield stress: 80 kPa at 250 kA/m - Off-state viscosity: 0.28 Pa·s at 40°C

Bingham plastic model:

$$\tau = \tau_y(H) + \eta_p \dot{\gamma}$$

At 0 field (driving, relaxed): $\tau_y = 0$ □ knob moves freely with no tactile resistance At 250 kA/m field (sport mode): $\tau_y = 80$ kPa □ firm, precise shifting feel

The electromagnet coil powering the field is a single ceramic-coated copper winding (electrically isolated from chassis — GIC barrier maintained via optical trigger signal).

3. YInMn Blue Ceramic Cap

The top contact surface is a sintered YInMn Blue glazed ZrO₂ ceramic disc: - Dielectric breakdown: 18 kV/mm — completely arc-repellent (GIC arc flashover prevented) - Thermal conductivity: 3 W/m·K — warm, comfortable hand feel (not cold like metal) - Hardness: 12 GPa — scratch and wear resistant

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